

Summer Bootcamp – Session 2

Hands-On Calculus and ODE Exercises

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1 Python and Elementary Financial Calculations

This first part requires only short Python functions and elementary algebra.

Exercise 1 – Coupon-bond yield, duration, and repricing. A bond pays 4 at $t = 1$, 4 at $t = 2$, and 104 at $t = 3$. Its market price is 96.50. We denote by $B(y)$ the price of the bond as a function of the continuously compounded yield y . Thus

$$B(y) = 4e^{-y} + 4e^{-2y} + 104e^{-3y}.$$

1. write Python functions for $B(y)$, $B'(y)$, and $B''(y)$;
2. find the yield with bisection on an interval whose endpoint prices bracket 96.50;
3. verify the result with Newton's method;
4. compute modified duration and convexity,

$$D_{\text{mod}} = -\frac{B'(y)}{B(y)}, \quad \mathcal{C} = \frac{B''(y)}{B(y)};$$

5. estimate the new price after yield moves of ± 25 and ± 100 basis points using duration alone and then duration-convexity;
6. compare both approximations with exact repricing.

2 Explicit Derivatives

Exercise 2 – One-dimensional derivatives. Differentiate and simplify each function. State its domain.

$$f_1(x) = \sin(3x^2 - 1), \quad f_2(x) = x^2 e^{-3x},$$

$$f_3(x) = \frac{\log x}{1 + x^2}, \quad f_4(x) = \log(1 + e^x), \quad f_5(x) = \sqrt{1 + x^2}.$$

For f_2 and f_4 , compute the second derivative as well.

Exercise 3 – Gradient, Hessian, and Jacobian. Let

$$f(x, y) = 2x^2 - xy + 3y^2 + 4x, \quad g(x, y) = \begin{pmatrix} e^x \cos y \\ x - y^2 \end{pmatrix}.$$

Using the method of your choice, compute symbolically:

1. the gradient $\nabla f(x, y)$;
2. the Hessian $H_f(x, y)$;
3. the Jacobian $J_g(x, y)$.

Exercise 4 – Multivariate chain rule by two methods. Define

$$q(u, v) = ue^v, \quad \Phi(x, y) = (x + y^2, xe^y), \quad F = q \circ \Phi.$$

1. write $F(x, y)$ explicitly and compute $\partial_x F$ and $\partial_y F$;
2. compute $J_\Phi(x, y)$ and $\nabla q(u, v)$;
3. recover ∇F from $\nabla F = J_\Phi^\top \nabla q(\Phi)$;
4. write the increment calculation $DF(x, y)h = Dq(\Phi(x, y))D\Phi(x, y)h$ and verify that it gives the same coefficients of h_1 and h_2 .

3 Explicit Integration

Exercise 5 – One-dimensional integrals. Evaluate analytically and state the method used:

$$I_1 = \int_0^1 (3x^2 - 2x + 1) dx, \quad I_2 = \int_0^T (1+t)^{-r} dt,$$

$$I_3 = \int_0^1 xe^{x^2} dx, \quad I_4 = \int_0^1 x \log x dx, \quad I_5 = \int_0^1 \frac{1}{1+x^2} dx.$$

For I_2 , distinguish $r = 1$ from $r \neq 1$ and recover the exceptional case by a limit. For I_5 , use the substitution $x = \tan u$.

Exercise 6 – The Gaussian integral. Let

$$G = \int_{-\infty}^{\infty} e^{-x^2} dx.$$

The objective is to obtain the value of G by a formal calculation.

1. write G^2 as an integral over $\mathbb{R}^2$ using independent variables x and y ;
2. change to polar coordinates, stating the domain and the Jacobian;
3. evaluate the radial and angular integrals and deduce G ;
4. use the result to compute $\int_{-\infty}^{\infty} e^{-x^2/(2\sigma^2)} dx$ for $\sigma > 0$;
5. explain how this calculation determines the normalizing constant of a Gaussian density.

Exercise 7 – Multiple integrals and change of variables. Compute

$$J_1 = \int_{-1}^2 \int_0^3 (2x - y) dy dx,$$

$$J_2 = \iint_D (x + 2y) dx dy, \quad D = \{(x, y) : x \geq 0, y \geq 0, x + 2y \leq 2\},$$

and

$$J_3(R) = \iint_{1 \leq x^2 + y^2 \leq R^2} (x^2 + y^2)^2 dx dy, \quad R > 1.$$

1. compute J_1 in both integration orders;
2. write both possible iterated-integral descriptions of D and compute J_2 ;
3. transform $J_3(R)$ to polar coordinates, including the Jacobian r ;
4. specialize the result to $R = 3$;
5. check the three answers with a short numerical quadrature in Python.

4 Black–Scholes Greeks

For a non-dividend-paying asset,

$$C = S\Phi(d_1) - Ke^{-rT}\Phi(d_2), \quad d_1 = \frac{\log(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}, \quad d_2 = d_1 - \sigma\sqrt{T}.$$

Here Φ is the cumulative distribution function of a standard normal random variable:

$$\Phi(x) = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^x e^{-u^2/2} du, \quad \phi(x) = \Phi'(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2}.$$

Exercise 8 – Derive the Greeks. The closed-form formulas for the Greeks are deliberately not supplied: they must be obtained by differentiating the Black–Scholes price.

With all other variables held fixed, the Greeks are defined by the following partial derivatives:

$$\Delta = \frac{\partial C}{\partial S}, \quad \Gamma = \frac{\partial^2 C}{\partial S^2}, \quad \text{Vega} = \frac{\partial C}{\partial \sigma}.$$

$$\Theta = -\frac{\partial C}{\partial T}, \quad \text{Rho} = \frac{\partial C}{\partial r}.$$

Here T denotes time to maturity, so calendar-time Theta is $-\partial C/\partial T$.

2. derive a useful identity relating $S\phi(d_1)$ and $Ke^{-rT}\phi(d_2)$;
3. use the chain rule to derive the closed-form formulas for all five Greeks, simplifying with the identity from the previous question;
4. Derive the short-maturity at-the-money approximation as follows.
 - (a) Use the at-the-money-forward convention

$$K = Se^{rT},$$

and explain why it is equivalent to equality between the spot price S and the discounted strike Ke^{-rT} .

- (b) Substitute this relation into d_1 and d_2 and prove that

$$d_1 = \frac{\sigma\sqrt{T}}{2}, \quad d_2 = -\frac{\sigma\sqrt{T}}{2}.$$

- (c) Rewrite the call price in the form

$$C = S \left[\Phi\left(\frac{\sigma\sqrt{T}}{2}\right) - \Phi\left(-\frac{\sigma\sqrt{T}}{2}\right) \right],$$

then use the symmetry $\Phi(-z) = 1 - \Phi(z)$.

- (d) Using the expansion

$$\Phi(z) = \frac{1}{2} + \frac{z}{\sqrt{2\pi}} + O(z^3) \quad (z \rightarrow 0),$$

show that, as $T \downarrow 0$,

$$C = S\sigma\sqrt{\frac{T}{2\pi}} + O(T^{3/2}).$$

- (e) Interpret the leading term: describe separately the effects of S , σ , and T , and explain why the short-maturity price is of order $\sqrt{T}$ rather than order T .

5 Numerical Schemes for ODEs

For $y' = f(t, y)$ on $t_n = nh$, recall

$$y_{n+1} = y_n + hf(t_n, y_n) \quad \text{and} \quad y_{n+1} = y_n + hf(t_{n+1}, y_{n+1}).$$

Exercise 9 – Euler accuracy and the classical inequalities. For $y' = -2y$, $y(0) = 1$, on $[0, 3]$:

1. derive explicit and implicit Euler updates and their closed forms;
2. implement both schemes for $h = 0.5, 0.25, 0.125, 0.0625$;
3. compute $\max_n |y(t_n) - y_n|$ using $y(t) = e^{-2t}$ and estimate the convergence order;
4. derive the explicit stability inequality $|1 - 2h| < 1$ and the corresponding restriction on h ;
5. verify that the implicit update multiplier $1/(1 + 2h)$ has modulus below one for every $h > 0$;
6. explain why stability does not imply accuracy for a very large step.

6 Solving Scalar and Multidimensional ODEs

Exercise 10 – Scalar linear equations. Solve exactly and verify by differentiation:

$$\begin{aligned} y' + 4y &= 2t, & y(0) &= -1, \\ y' - \frac{1}{1+t}y &= (1+t)^2, & y(0) &= 2, \\ y' - y &= e^{2t}, & y(0) &= 0. \end{aligned}$$

Use variation of the constant for the first and third equations and an integrating factor for the second.

Exercise 11 – Nonlinear scalar equations. For an autonomous ODE $y' = f(y)$, an *equilibrium* is a constant solution $y(t) \equiv y^*$, equivalently a value satisfying $f(y^*) = 0$.

Solve the following equations analytically whenever possible:

$$\begin{aligned} y' &= y(1 - y), & y(0) &= y_0 \in (0, 1), \\ y' &= -y^3, & y(0) &= 1, \\ y' &= t(1 + y^2), & y(0) &= 0. \end{aligned}$$

1. For $y' = y(1 - y)$, first identify the equilibrium solutions. For a non-equilibrium solution, divide by $y(1 - y)$ and use

$$\frac{1}{y(1 - y)} = \frac{1}{y} + \frac{1}{1 - y}.$$

Integrate both sides, determine the integration constant from $y(0) = y_0$, then solve explicitly for $y(t)$.

2. For $y' = -y^3$, rewrite the equation with all powers of y on the left. Integrate, use $y(0) = 1$, and choose the branch consistent with the initial condition. Verify the answer by differentiation.
3. For $y' = t(1 + y^2)$, separate the variables and recognize an elementary antiderivative on the left. After applying $y(0) = 0$, solve for $y(t)$ and inspect where that expression ceases to be finite.
4. For each solution, determine its maximal interval of existence. Do not infer this interval only from the ODE: inspect the explicit denominator or the inverse function introduced during the calculation.
5. Draw the phase line for the two autonomous equations and determine the stability of every equilibrium.

Exercise 12 – Linear systems in dimension two. Consider

$$x'(t) = Ax(t), \quad A = \begin{pmatrix} -2 & 1 \\ 0 & -4 \end{pmatrix}, \quad x(0) = \begin{pmatrix} 1 \\ 2 \end{pmatrix}.$$

1. compute the eigenvalues and eigenvectors of A ;
2. decompose $x(0)$ into eigenvectors and obtain the exact solution;
3. verify the solution using $x(t) = e^{tA}x(0)$;
4. implement explicit and implicit Euler in vector form;
5. compare both numerical trajectories with the exact solution.

Exercise 13 – Forced multidimensional system. Let

$$x' = Ax + b, \quad A = \begin{pmatrix} -2 & 0 \\ 1 & -1 \end{pmatrix}, \quad b = \begin{pmatrix} 2 \\ 0 \end{pmatrix}, \quad x(0) = \begin{pmatrix} 0 \\ 2 \end{pmatrix}.$$

1. compute the equilibrium $x^* = -A^{-1}b$;
2. use $x(t) = x^* + e^{tA}(x(0) - x^*)$ to obtain an exact solution;
3. derive the same answer component by component;
4. implement Euler schemes and verify convergence to x^* .

Exercise 14 – Second-order equation as a system. Solve

$$y'' + 4y' + 3y = 0, \quad y(0) = 0, \quad y'(0) = 2.$$

1. use the characteristic polynomial;
2. rewrite the equation as a first-order system in (y, y') ;
3. compute the eigenvalues of the companion matrix;
4. implement explicit Euler for the system and compare with the exact solution obtained in the first question.